

Matthew Snelgrove

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matthewSnelgrove
portfolio

Education

University of Toronto, Scarborough | Computer Science | CGPA: 3.72 Sept. 2021 – Present

- Courses: Data Structures, Algorithms, Software Design, Security, Databases, Web Development

Experience

SS&C Technologies (IFDS) | Software Developer September. 2023 – April 2024

- Redesigned auditing app into modern webapp with Node and React leveraging internal APIs to replace expensive file parsing and decrease load times by 99%
- Documented codebase with jsDoc and created comprehensive wiki entries minimizing maintenance costs and expediting project handover
- Identified and patched production bugs across team apps rapidly, minimizing impact on business operations
- Investigated new internal tools and created tailored user guides to facilitate effective integration into core apps

Ontario Ministry of Transportation | Junior Technical Analyst January. 2023 – April 2023

- Developed RESTful API with Java Spring Boot for modernization of critical driver/vehicle registration systems
- Performed automated unit and integration testing with JUnit and Postman to verify functionality and identify errors caused by frequently changing internal dependencies, reducing debugging time by over 50%
- Ran builds and deployments with Maven and Azure Pipelines to eliminate time consuming and error-prone manual builds and save ~3 hours per week of developer time

Projects

Superbiddo   October 2024 – December 2024

- Developed flexible and scalable RESTful API using Express with comprehensive documentation following OpenAPI 3.0 specification to enable straightforward integration by client
- Integrated Google Cloud Storage to remotely store and retrieve large image files, reducing load on the server, and added lifecycle rules to delete orphaned files automatically, reducing bloat and cloud costs
- Deployed app using three-tier architecture with HTTPS on a Google Cloud VM, creating Docker containers for the frontend, backend, and database layers to enable targeted optimization, robust scalability, and independent updates to layers
- Configured Google Cloud VM instance to host containerized application, ensuring efficient resource utilization, minimal cost for performance requirements, and secure networking
- Automated deployments with bash script to build+upload images and start containers with secrets using Docker Compose file, saving hours of manual deployment time and ensuring consistent, reliable deployments
- Secured app with helmet.js, data validation, CSRF protection, and session-based authentication, fully protecting the app from exploitation and earning a bonus during grading

Stock Portfolio App  August 2024

- Designed and implemented database for social stock tracking platform
- Optimized statistical correlation and covariance queries by precomputing results in a materialized view, reducing query time by 98%

Technical Skills

Languages: Java, JavaScript/TypeScript, Python, C/C++, C#, SQL (Postgres), HTML, CSS

Frameworks & Libraries: Node.js, Express.js, Spring Boot, React, MongoDB, JUnit, jQuery

Developer Tools: Git, Docker, Google Cloud Platform, Azure, Postman, OpenApi